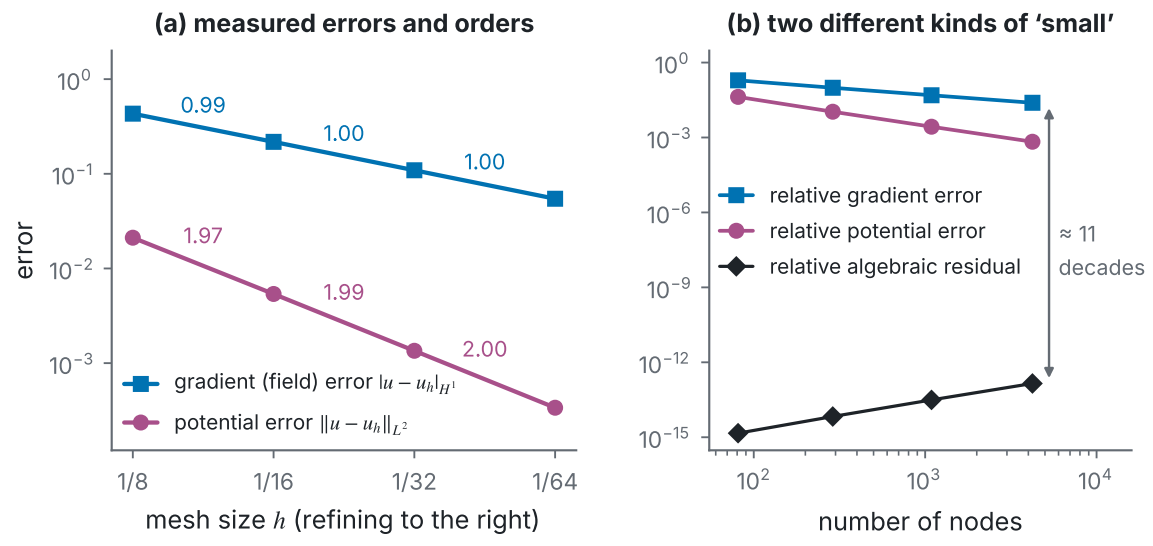
(a) measured errors and orders
error
10^{0}
10^{-1}
10^{-2}
10^{-3}
0.99
1.00
1.00
1.97
1.99
2.00
gradient (field) error $|u - u_h|_{H^1}$
potential error $\|u - u_h\|_{L^2}$
1/8
1/16
1/32
1/64
mesh size h (refining to the right)
(b) two different kinds of 'small'
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
relative gradient error
relative potential error
relative algebraic residual
≈ 11
decades
10^{2}
10^{3}
10^{4}
number of nodes